THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · EASY LEVEL CHAPTER TEST SET 1

Chapter 14: **Surface Area and Volume** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The volume of a cuboid of length 4 cm, breadth 8 cm and height 7 cm is: 1

(a) 201 cm^3 **(b)** 270 cm^3 **(c)** 247 cm^3 **(d)** 224 cm^3

2. The volume of a cube of edge 5 cm is: 1

(a) 112 cm^3 **(b)** 152 cm^3 **(c)** 138 cm^3 **(d)** 125 cm^3

3. The total surface area of a cube of edge 4 cm is: 1

(a) 117 cm^2 **(b)** 106 cm^2 **(c)** 86 cm^2 **(d)** 96 cm^2

4. The lateral surface area of a cuboid 9 cm $\times$ 5 cm $\times$ 3 cm (height 3 cm) is: 1

(a) 75 cm^2 **(b)** 84 cm^2 **(c)** 102 cm^2 **(d)** 93 cm^2

5. The volume of a cone is ___ the volume of a cylinder of the same base and height: 1

(a) equal to **(b)** twice **(c)** one-third of **(d)** half of

6. The volume of a cube is 8 cm^3. Its edge is: 1

(a) 1 cm **(b)** 4 cm **(c)** 2 cm **(d)** 3 cm

7. Take $\pi = \frac{22}{7}$. The volume of a cylinder of radius 7 cm and height 3 cm is: 1

(a) 556 cm^3 **(b)** 462 cm^3 **(c)** 415 cm^3 **(d)** 509 cm^3

8. Take $\pi = \frac{22}{7}$. The surface area of a sphere of radius 21 cm is: 1

(a) 6654 cm^2 **(b)** 5544 cm^2 **(c)** 6099 cm^2 **(d)** 4989 cm^2

9. The volume of a cuboid of length 3 cm, breadth 9 cm and height 6 cm is: 1

(a) 196 cm^3 **(b)** 179 cm^3 **(c)** 162 cm^3 **(d)** 145 cm^3

10. The volume of a cube of edge 7 cm is: 1

(a) 413 cm^3 **(b)** 308 cm^3 **(c)** 343 cm^3 **(d)** 378 cm^3

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the volume, total surface area and lateral surface area of a cuboid of length 9 cm, breadth 6 cm and height 3 cm. 2

12. Find the volume and total surface area of a cube of edge 5 cm. Also find the edge of a cube whose volume is 216 cm³. 2

13. A water tank is 8 m long, 4 m wide and 7 m deep. How many litres of water does it hold when full? (1 m³ = 1000 litres) 2

14. Find the volume, total surface area and lateral surface area of a cuboid of length 8 cm, breadth 7 cm and height 2 cm. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. Find the volume, total surface area and lateral surface area of a cuboid of length 5 cm, breadth 15 cm and height 9 cm. 3

16. Take $\pi = \frac{22}{7}$. Find the curved surface area, total surface area and volume of a cylinder of radius 3.5 cm and height 11 cm. 3

17. Find the volume and total surface area of a cube of edge 17 cm. Also find the edge of a cube whose volume is 5832 cm³. 3

OR

Find the volume, total surface area and lateral surface area of a cuboid of length 5 cm, breadth 12 cm and height 13 cm. 3

18. Take $\pi = \frac{22}{7}$. Find the curved surface area, total surface area and volume of a cylinder of radius 3.5 cm and height 7 cm. 3

OR

Find the volume and total surface area of a cube of edge 16 cm. Also find the edge of a cube whose volume is 4913 cm³. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Take $\pi = \frac{22}{7}$. Find the curved surface area, total surface area and volume of a cylinder of radius 21 cm and height 5 cm. 5

OR

How many cubes of edge 5 cm can be cut from a cuboid measuring 15 cm × 30 cm × 25 cm? Find the leftover volume if each cube is cut and the cuboid is fully used. 5

20. Take $\pi = \frac{22}{7}$. Find the curved surface area, total surface area and volume of a cylinder of radius 7 cm and height 16 cm. 5

OR

How many cubes of edge 4 cm can be cut from a cuboid measuring 20 cm × 8 cm × 12 cm? Find the leftover volume if each cube is cut and the cuboid is fully used. 5

END OF QUESTION PAPER Total: 40 Marks

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